

Sokoban Algorithm Performance Report

Algorithm Performance Comparison by Level

Time (s) Comparison

Level	SwarmAlgorithm	BFS	UCS	AStar	Dijkstra	GreedyBFS	DFS
1	0.11	0.04	0.13	0.06	0.14	0.00	0.00
2	0.02	0.00	0.00	0.00	0.00	0.00	0.00
3	0.03	0.00	0.01	0.01	0.01	0.00	0.00
4	0.03	0.00	0.00	0.00	0.00	0.00	0.00
5	0.57	0.35	0.40	0.48	0.40	0.09	0.15
6	0.19	0.18	0.18	0.09	0.18	0.06	0.15
7	2.71	0.38	0.48	0.69	0.49	0.10	0.05
8	566.67	77.58	47.03	0.46	51.56	0.00	13.15
9	190.47	9.12	141.05	17.19	140.86	11.10	N/A
10	N/A	32.54	19.23	61.19	18.75	47.78	10.65

Nodes Expanded Comparison

Level	SwarmAlgorithm	BFS	UCS	AStar	Dijkstra	GreedyBFS	DFS
1	7981	8845	30720	9320	30604	328	586
2	406	405	406	405	406	305	304
3	1679	1734	1910	1775	1893	1033	715
4	975	989	1214	930	1192	201	654
5	26967	88889	92105	68883	91788	15001	45846
6	10933	52144	43086	13089	43014	10565	41746
7	65763	97309	106911	84012	106481	13815	12074
8	724108	2363648	4370538	45199	4364480	349	2204290
9	473688	526939	9961059	1310132	9941588	1049406	N/A
10	N/A	1907416	1912293	1885153	1912295	1905343	1899680

Memory Usage (MB) Comparison

Level	SwarmAlgorithm	BFS	UCS	AStar	Dijkstra	GreedyBFS	DFS
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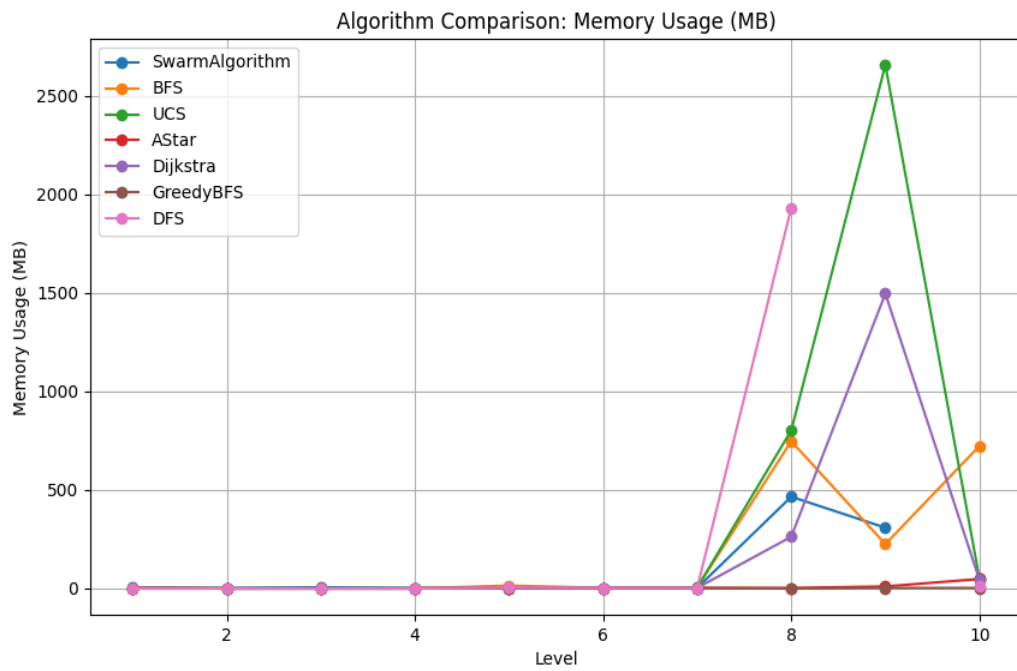
1	4.41	3.03	3.03	3.03	3.03	0.00	0.00
2	3.05	0.00	0.00	0.00	0.00	0.00	0.00
3	6.08	0.00	0.00	0.00	0.00	0.00	0.00
4	3.12	0.00	0.00	0.00	0.00	0.00	0.00
5	3.12	15.23	3.03	3.03	3.03	3.03	4.55
6	3.03	3.03	3.03	3.03	3.03	3.03	3.03
7	3.03	5.25	4.62	3.03	3.03	3.05	3.03
8	465.64	747.39	802.42	3.03	262.70	0.00	1931.17
9	308.69	225.30	2656.53	10.61	1496.22	3.03	N/A
10	N/A	723.52	49.59	48.39	44.08	2.88	12.06

Solution Steps Comparison

Level	SwarmAlgorithm	BFS	UCS	AStar	Dijkstra	GreedyBFS	DFS
1	16	15	24	26	24	28	86
2	71	70	71	71	71	71	71
3	61	60	65	67	65	67	281
4	69	68	75	75	75	81	217
5	121	120	125	125	125	255	2311
6	112	111	140	142	140	278	2988
7	80	79	84	80	84	92	146
8	38	37	46	38	46	57	7445
9	31	30	47	43	47	92	N/A
10	N/A	58	59	59	59	59	307

Total Weight Comparison

Level	SwarmAlgorithm	BFS	UCS	AStar	Dijkstra	GreedyBFS	DFS
1	695	596	405	405	405	405	803
2	72	66	72	72	72	72	72
3	46	42	40	38	40	38	90
4	48	46	34	34	34	56	118
5	203	198	181	181	181	308	1949
6	1923	1749	1611	1609	1611	4149	33543
7	153	149	149	153	149	153	185
8	55	54	582	590	582	777	35521



Statistical Summary

Metric	SwarmAlgorithm	BFS	UCS	AStar	Dijkstra	GreedyBFS	DFS
Avg Time (s)	84.53	12.02	20.85	8.02	21.24	5.92	2.68
Avg Nodes	145833	504831	1652024	341889	1649374	299634	467321
Avg Memory (MB)	88.91	172.28	352.23	7.42	181.51	1.50	217.09
Avg Steps	66	64	73	72	73	108	1539
Avg Weight	462	447	459	460	459	816	8298